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Two million years of flaking stone and the evolutionary efficiency of stone tool technology

Željko Režek ^{1,2*}, Harold L. Dibble^{1,3,4}, Shannon P. McPherron ¹, David R. Braun^{1,5} and Sam C. Lin⁶

¹Department of Human Evolution, Max Planck Institute for Evolutionary Anthropology, Leipzig, Germany. ²University of Pennsylvania Museum of Archaeology and Anthropology, Philadelphia, PA, USA. ³Department of Anthropology, University of Pennsylvania, Philadelphia, PA, USA. ⁴Institute of Human Origins, School of Human Evolution and Social Change, Arizona State University, Tempe, AZ, USA. ⁵Department of Anthropology, George Washington University, Washington DC, USA. ⁶Centre for Archaeological Science, School of Earth and Environmental Sciences, University of Wollongong, Wollongong, Australia. *e-mail: zeljko_rezek@eva.mpg.de

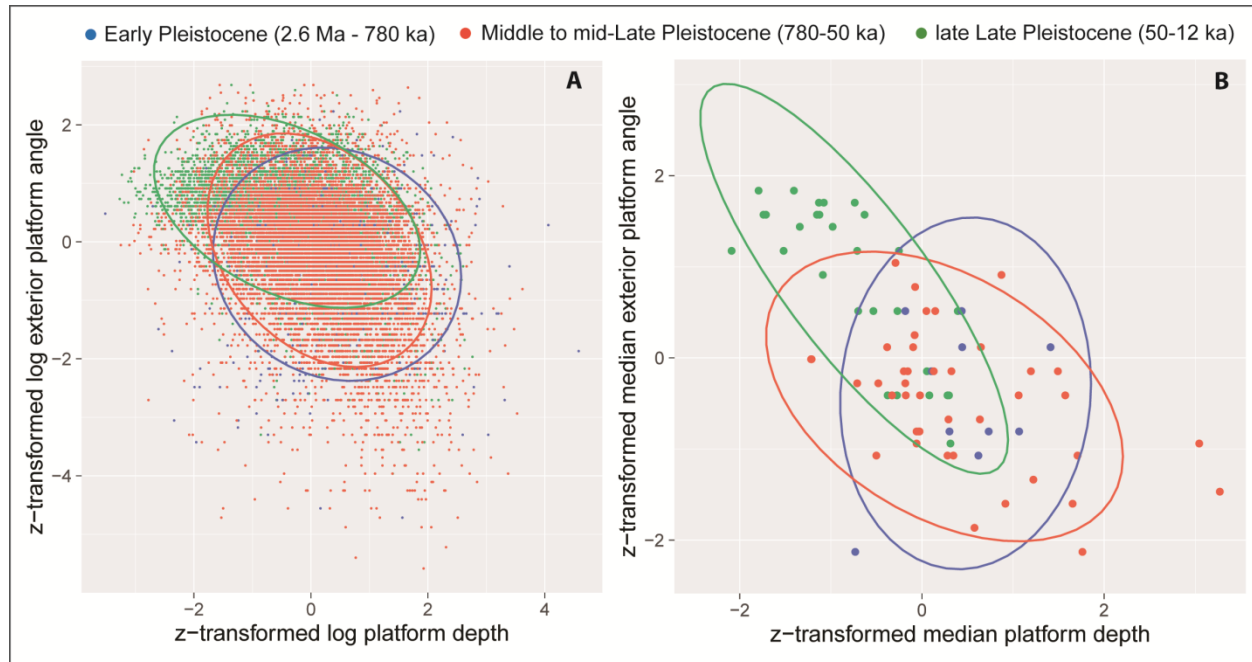
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Supplementary Information

Distribution of assemblages in Figure 1.

The Acheulian assemblage from Elandsfontein ¹ (“11”) in South Africa, some Quina Mousterian assemblages from Combe-Capelle Bas ² (“39”-“42”) and Quina Mousterian assemblages from Roc de Marsal ³ (“32”-“34”) in France exhibit relatively high median values of platform depth and relatively low median values of exterior platform angle, indicating production of flakes with low amounts of sharp edges. The Early Stone Age assemblages FwJj20 ⁴ (“1”) and FxJj20M ⁵ (“9”) from Kenya, the Mousterian assemblages from Bisitun ^{6,7} (“18”) and Warwasi ⁸ (“19”) in Iran, the Levantine Mousterian assemblages from Skhul ⁹ (“15”), Tabun ¹⁰ (“12”), and Rosh Ein Mor ¹¹ (“13” and “14”) and Ahmarian assemblages from Boker ¹² (“60”) and Ein Aqev East ¹² (“63”) in Israel, the Châtelperronian assemblages from La Côte ¹³ (“53”) and Roc de Combe ^{14,15} (“54”) in France, two Recent Aurignacian assemblages ¹⁶ (“65” and “66”) and Gravettian assemblages ¹⁷⁻¹⁹ (“67”-“70”) from Abri Pataud in France, Gravettian, Solutrean, Badegoulian, and Magdalenian assemblages from Laugerie Haute Est ²⁰⁻²³ (“71”-“77”, “79”, “80”), Fourneau du Diable ^{23,24} (“78”) and La Madeleine ²³ (“81”) in France, and Khormusan assemblage 1017 ²⁵ (“24”) from northern Sudan exhibit low median values of platform depth and high median values of exterior platform angle, indicating high production of sharp edges. The smallest flakes are from the Oldowan assemblage FxJj18G ⁵ (“8”) from Kenya; the largest are from the Levantine Mousterian assemblage D40 ²⁶ (“25”) from the Negev. Flakes from the rest of Quina Mousterian assemblages from Combe-Capelle ² (“43”-“46”) and the Oldowan assemblage FxJj38NW ⁵ (“5”) are between the biggest flakes and those with small amounts of sharp edge. In between all these assemblages are the rest of the Early Stone Age assemblages from Kenya ^{5,27} (“2”-“4”, “6”, “7”, “10”), the Middle Stone Age assemblages from Contrebandiers Cave ²⁸ (“16” and “17”) in Morocco, a Levantine Mousterian assemblage from Ksar Akil ²⁹ (“37”) in Lebanon, the rest of Mousterian assemblages from Roc de Marsal ³ (“26”-“30”), Mousterian assemblages from Abri Peyrony ³⁰ (“47”-“50”) and Pech de l’Azé IV ³¹⁻³³ (“20”-“23”, “31”, “35”, “36”, “51”, “52”), a Mousterian assemblage from Combe Capelle Bas (“38”), the Early Aurignacian ¹⁶ (“55”-“59”) and one Recent Aurignacian ¹⁶ (“64”) assemblages from Abri Pataud, and the Levantine Aurignacian assemblages from Ein Aqev ¹² (“61” and “62”) in Israel.

Supplementary Figure 1. Distributions (95% confidence ellipses) of values of platform depth and exterior platform angle of individual flakes (A) and assemblages (B) of the three chronological groups.



Supplementary Table 1. The median and the interquartile range in the amount of sharp edge among all sampled complete and unmodified flakes within three time periods.

Period	N	Sharp edge amount	
		Median	Interquartile range
2 Ma – 780 ka	867	13.16	13.8
780 – 50 ka	11,907	22.74	24.99
50 – 12 ka	5,347	27.04	31.16

Supplementary Table 2. Linear regression between stone transport, edge or tool maintenance, and sharp edge production in the Neanderthal assemblages from southwest France.

DFFit=0.78; DFBeta edge/tool maintenance orig. -0.106, min. -0.13, max. -0.085;
 DFBeta stone transport orig. -0.158, min. -0.179, max. -0.135; Cook's distance 0.17;
 Leverage 0.24 (threshold 0.31); VIF 1.27

Edge/tool maintenance Estimate=-0.105, SE=0.052, t=-2.019, p=0.055, lower CI=-0.214, upper CI=0.003
 Stone transport Estimate=-0.158, SE=0.052, t=-3.029, p=0.006, lower CI=-0.266, upper CI=-0.05
 F=11.94, df=2,23, p<0.001; Adjusted R-squared 0.47;
 R-squared for edge/tool maintenance 0.15; R-squared for stone transport 0.29

Assemblages (N=26):
 Roc de Marsal layers 2, 3, 4, 5, 6, 7, 8, 9;
 Combe-Capelle Bas layers I-2B, I-2A, I-1E, I-1D1, I-1D, II-4A, II-4B, II-4C, II-4;
 Pech de l'Azé IV layers 8, 6B, 6A, 5B, 5A, 4C, 4A, 3B, 3A.

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